

Dataset Construction Plan

Final Planning Gate Before Any Dataset Generation

Codex-assisted surrogate planning report

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Abstract

This report converts the approved surrogate dataset design into exact future construction instructions. It approves construction planning only. It does not generate a dataset, solve the planned regimes, create `.npz`, `.npy`, or `.pt` files, export labels, create training tables, create model files, train neural networks, or run broad production-scale sweeps. The future label source remains the validated baseline American Crank–Nicolson / projected SOR solver.

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1 Purpose of Dataset Construction Plan

The surrogate dataset design plan approved only a schema and policy. This construction plan is the next gate: it names the planned regimes, dry-run subset, split assignment, future storage manifest, failure handling policy, and approval checklist. It is still a planning document. No regime is solved here and no label array is produced.

The construction objective is a future dataset for constrained surrogate experiments on American option value and continuation premium, with boundary and Greek diagnostics preserved as metadata or masked labels. The project remains grounded in the validated finite-difference American option benchmark and free-boundary setting (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Design Plan Versus Construction Plan

Artifact	What it decides	What it does not do
Dataset design plan	Feature schema, label schema, parameter grid, masks, diagnostics, and split policy.	It does not name every regime or authorize generation.
Construction plan	Exact regime list, dry-run subset, split assignment, future storage manifest, and failure rules.	It does not run the solver or create dataset files.
Future dry run	A small future execution over eight regimes after approval.	It is not performed in this report.
Future small dataset	A later <code>v1_small_grid</code> generation if dry-run review passes.	It remains blocked until human review.

3 Source Evidence and Previous Gates

The construction plan inherits constraints from the solver-validation synthesis, application strength gate, downstream planning, Pilot 01, quantitative risk analytics, and the surrogate dataset design plan. The methodology handouts warn that neural surrogates should be trained only after the classical benchmark and application diagnostics are credible. They also warn against random-only splits and unmasked Gamma labels near payoff kinks and free-boundary regions.

Prior source	Construction constraint
Ticket 12 synthesis	Solver validation passed with limitations, not production certification.
Application-strength gate	Dataset planning is allowed; dataset construction remains blocked.
Dataset design plan	Use normalized $K = 1$, baseline CN/PSOR, planned feature and label schemas, masks, diagnostics, and regime-level splits.
Methodology handouts	Use validated PSOR labels, stress regimes, held-out regimes, and boundary/Gamma caution.

4 Planned Full Regime Grid and Count

The future full small-grid dataset is planned as the Cartesian product:

$$\begin{aligned} \text{option_type} &\in \{\text{put}, \text{call}\}, & T &\in \{0.25, 0.5, 1.0, 2.0\}, & \sigma &\in \{0.20, 0.40, 0.60\}, \\ r &\in \{0.01, 0.05, 0.10\}, & q &\in \{0.00, 0.03, 0.06, 0.10\}. \end{aligned}$$

This gives:

$$2 \times 4 \times 3 \times 3 \times 4 = 288$$

planned regimes: 144 American puts and 144 American calls. The 36 no-dividend call regimes are included as validation/control regimes. They are not core dividend-call early-exercise evidence.

Every planned regime uses:

$$K = 1, \quad S_{\max} = 4K, \quad M = N = 120, \\ \text{solver_variant} = \text{baseline_cn_psor}.$$

Regime-plan rule	Exact policy
Regime ID format	<code>\{option_T\{025 050 100 200\}_s\{020 040 060\}_r\{001 005 010\}_q\{000 003 006 010\}</code> .
Machine-readable plan	<code>reports/05_surrogate/dataset_regime_plan.csv</code> .
Generated now	No. The CSV is a plan, not solved data.
Control rows	Call rows with $q = 0$ are marked as no-dividend call controls.

5 Dry-Run Subset Plan

Before any 288-regime generation, the future implementation must run a small manual-review dry run. The dry run is planned as eight regimes:

Dry-run ID	Regime ID	Purpose
dry_01	put_T100_s020_r005_q003	Base-like American put.
dry_02	put_T100_s060_r005_q003	High-volatility put.
dry_03	put_T100_s020_r001_q003	Low-rate put.
dry_04	put_T200_s040_r010_q010	Longer, higher-rate stress put.
dry_05	call_T100_s020_r005_q006	Base-like dividend-paying call.
dry_06	call_T100_s020_r005_q010	High-dividend call.
dry_07	call_T100_s060_r005_q006	High-volatility dividend call.

Dry-run ID	Regime ID	Purpose
dry_08	call_T100_s020_r005_q000	No-dividend call control.

The machine-readable dry-run plan is listed in the reproducibility and source record. It is not a dry-run result table.

6 Feature and Label Schema Summary

The future feature vector remains:

$$x = (\log(S/K), \tau/T, r, q, \sigma, T, \text{is_call}),$$

with solver and grid metadata retained separately. The main learning and reporting region remains:

$$S/K \in [0.4, 1.8], \quad \tau/T \in [0, 1].$$

The most reliable future labels are **value_over_K** with formula U/K , **payoff_over_K** with formula Φ/K , and **premium_over_K** with formula $(U - \Phi)/K$. Exercise indicators and boundary locations are threshold-based diagnostics. Delta is cautioned. Scaled Gamma is high caution and must never be used unmasked near payoff kinks, maturity rows, or extracted boundary bands.

7 Mask and Diagnostic Policy Summary

Future generated surfaces must preserve masks for payoff-kink-near, boundary-near, maturity-row, strict-interior, Gamma-allowed, Delta-allowed, exercise-region, and continuation-region nodes. Masks are mandatory metadata, not decoration.

Every future solved regime must record:

- solver convergence flag;
- maximum and mean PSOR iterations;
- maximum final PSOR update;
- maximum obstacle violation;
- maximum equation violation;
- maximum absolute complementarity product;
- boundary found count and boundary threshold;
- grid metadata $M, N, S_{\max}, \Delta S, \Delta \tau$;
- solver name and solver variant;
- downstream use status.

8 Split Assignment Plan

The split is regime-level, not row-level and not random point only. The deterministic precedence is:

1. **stress_holdout**: $\sigma = 0.60$ and $q = 0.10$;
2. **test**: all call regimes with $q = 0.00$, plus $T = 2.0$, $r = 0.10$, $\sigma = 0.40$, after stress precedence;
3. **validation**: $T = 0.5$ and $r = 0.05$, after stress/test precedence;
4. **train**: all remaining regimes.

Split	Planned count	Purpose
train	202	Main fitting regimes for future surrogate experiments.
validation	19	Model-selection regimes after stress/test precedence.
test	43	No-dividend call controls and selected long-maturity high-rate regimes.
stress_holdout	24	High-volatility, high-dividend stress regimes.

The machine-readable assignment is listed in the reproducibility and source record. No generated rows exist yet.

9 Storage and Manifest Plan

The future storage package may use one compressed `.npz` numeric-array file plus CSV or JSON manifests. This report does not create that storage. The planned storage manifest has `created_now=no` for every row.

Future file	Type	Purpose
Future arrays file	compressed arrays	Future numeric samples only after approval; exact <code>.npz</code> name is in the CSV plan.
Regime manifest	CSV manifest	Regime parameters and solver settings.
Diagnostic summary	CSV manifest	PSOR, LCP, boundary, runtime, and mask diagnostics.
Split assignment	CSV manifest	Approved regime-level split assignment.
Label schema copy	CSV schema copy	Frozen future label schema.
Mask policy	JSON manifest	Exact mask thresholds and widths.
Generation log	text log	Future command, environment, and failure log.
Validation report	PDF report	Future post-generation review report.

10 Failure Handling Policy

The future generation script must reject or quarantine regimes according to the failure policy before any label export:

Failure type	Detection rule	Default action
PSOR non-convergence	PSOR convergence flag is false.	Exclude and review.
Obstacle violation	Obstacle violation exceeds 10^{-8} .	Exclude and review.
Equation violation	Equation violation exceeds 10^{-6} .	Exclude and review.
Complementarity failure	Complementarity product exceeds 10^{-6} .	Exclude and review.
Boundary not found	Boundary flag false or count unexpectedly low.	Keep price/premium only if other diagnostics pass; boundary label missing.
Greek masks missing	Required masks absent or unapproved.	Block Delta/Gamma labels.
Unexpected runtime	Runtime exceeds approved budget.	Mark expensive and review.
Missing metadata	Required provenance or diagnostics absent.	Block export until fixed.

11 Construction Approval Checklist

Actual dataset construction may start only after human review approves:

- exact regimes and 288-regime count;
- dry-run subset;
- split assignment;
- storage format;
- diagnostic thresholds;
- failure handling policy;
- mask widths;
- post-dry-run review process.

The checklist is listed in the reproducibility and source record. It records planning status only.

12 Blocked Actions

The following remain blocked by this plan:

- actual dataset construction;

- CN/PSOR solves over the planned regimes;
- `.npz`, `.npy`, or `.pt` file creation;
- label-array export;
- training-table export;
- model-file creation;
- neural surrogate training;
- broad production-scale sweeps;
- production Greek labels;
- final paper claims across all regimes.

13 Limitations

This construction plan is detailed enough to implement a future dry-run generator, but it is not evidence that the dataset exists. The split counts are deterministic and auditable, but they have not been tested on generated rows. Boundary and Gamma labels remain diagnostic and mask-dependent. The planned $M = N = 120$ grid is inherited from pilot defaults and should be confirmed on selected $M = N = 180$ cases before broad scaling.

14 Recommended Next Step

The recommended next step is human review of this construction plan and its six planning CSVs. If accepted, the next technical ticket should implement only the eight-regime `v0_dry_run` generator, parse the approved planning CSVs, write no full 288-regime dataset, and produce a post-dry-run review report before any `v1_small_grid` generation.

15 Reproducibility and Source Record

Item	Record
Regime plan	<code>reports/05_surrogate/dataset_regime_plan.csv</code> .
Dry-run subset	<code>reports/05_surrogate/dataset_dry_run_subset_plan.csv</code> .
Split assignment	<code>reports/05_surrogate/dataset_split_assignment_plan.csv</code> .
Storage manifest plan	<code>reports/05_surrogate/dataset_storage_manifest_plan.csv</code> .
Failure handling policy	<code>reports/05_surrogate/dataset_failure_handling_policy.csv</code> .
Construction checklist	<code>reports/05_surrogate/dataset_construction_checklist.csv</code> .

Item	Record
Construction backlog	reports/05_surrogate/dataset_construction_backlog.md.
Shared bibliography	reports/03_solver/references.bib.

References

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